Iowa State University
University Honors Program

Engineering Honors Project
Fall 2024

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Developing Team Leadership Strategies to Grow a Software Startup

Project Overview: Casmium, LLC

Casmium Sports Analytics provides convenient, affordable, and accessible tools to collect, store, and analyze baseball and softball stats for individuals and teams. Instead of using paper, teams can quickly and effortlessly collect powerful stats during games, then view sophisticated and personalized analytics, aggregate insights from multiple third-party tools, and export stats to popular league platforms all in a single dashboard.

As CEO of Casmium, my goal is simple: develop and distribute a working analytics solution to teams across the country and serve a need in the sports community. This semester, my project involved three pillars, Leadership, Entrepreneurship, and Engineering, to make progress on our long-term business goal with my team.

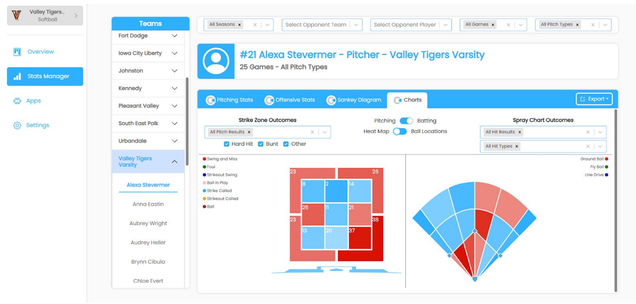


Figure 1: *Stats Dashboard* v0.3, with changes to the data selection and filtering UI requested from platform users

- **Leadership**: Improve team leadership skills and provide awareness to leadership concepts and frameworks (CORE and VUCA models) to develop team member talent
- **Entrepreneurship**: Attain 50 clients from any high school, fall ball/travel, little league team categories by Summer 2025
- **Engineering**: Release Stats Tracker (app) and Stats Dashboard (website) feature improvements and integrate with one 3rd-party tool by Winter 2025

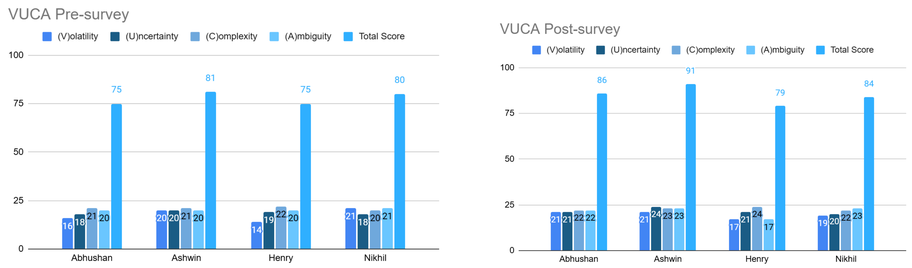
Leadership

CORE (Confidence, Optimism, Resilience, Engagement) Framework:

- Evaluation of Self: How do my internal skills reflect leadership?
- Success and positivity with confidence
- More possibilities observed with optimism
- Learning from failure, more engagement and resiliency
- A balance is necessary (confidence with humility, optimism with reality, resilience with willingness to change, engagement with balance)

VUCA (Volatility, Uncertainty, Complexity, Ambiguity) Framework:

- Evaluation of skills in a face-paced, changing business environment
- Our business needs team members with adaptability, resiliency, agility, and strategic thinking
- Startup landscape is unknown and rapidly changing
- VUCA measures team effectiveness against these



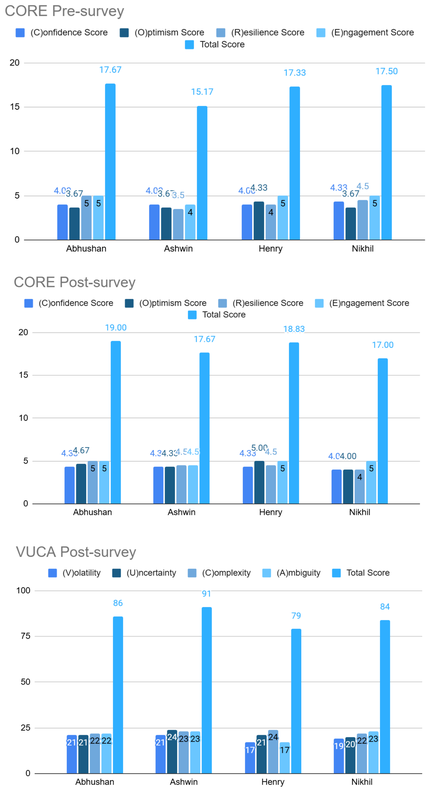


Figure 2: *CORE and VUCA Pre and Post-Surveys of the team's leadership traits*

Engineering

- Feedback from Valley Softball, Waukee Baseball
- Engineering decisions as a team to move towards simpler filter and selection of teams, players, season, and opponents when querying data
- See Figure 1 for updates to *Stats Dashboard* published
- See Figure 6 for updates to *Stats Tracker* set to publish before January
- Integration with *MaxPreps* to automate stats reporting to leagues
- Working towards local partnership with *Bound* in Ames

Entrepreneurship

- Online marketing vs offline marketing: Coaches like in-person or face-to-face interactions, see Figure 4
- Video demos, calls, and event meet-ups work well to convert. See Figure 3 for planned event locations in 2025
- Cold email, social media, and online marketing work less
- Online presence still important in sports world
- $4,500 to spend on events attendance, see Figure 5 budget

Project Achievements

- 2 customers since project start (9 all-time retained)
- Raised additional $2,500 in funding through pitch competitions
- Registered for 7 coach clinics
- Pursued podcast and social media marketing channels, 400 email contacts
- Updated dashboard and scoring app user interfaces based on user feedback
- Cloud computing infrastructure improvements (cost and efficiency)



Figure 3: Location of clinics Casmium will be attending



Figure 4: The Casmium team demoing product at the 2024 Iowa High School Baseball Coaches Clinic

	Name	Description	Unit	Estimated Cost	Purchased?	Link	Notes
2	**Media**						
3	Hawkeye of the Storm Podcast	2 30 second ad rolls	7 games * 2 ads	$700.00	☐	Link	See 2025 Events for Travel Costs
4	**Events**						
5	Minnesota HSBCA		1	$347.56	☑	Link	See 2025 Events for Travel Costs
6	Illinois HSBCA		1	$841.00	☑	Link	See 2025 Events for Travel Costs
7	Iowa HSBCA	Includes 8' table	1	$200.00	☐	Link	See 2025 Events for Travel Costs
8	Iowa DCA Softball	Includes 8' table	1	$50.00	☐	Link	See 2025 Events for Travel Costs
9	Missouri HSBCA		1	$723.64	☐	Link	See 2025 Events for Travel Costs; 2 PRIZES NEEDED!
10	Nebraska Baseball Coaches Association		1	$300.00	☐	Link	See 2025 Events for Travel Costs
11	**Marketing Materials**						
12	Foam Board Graphic	Behind table	2	$50.00	☐	Bipeline & Blue	Ingersoll, Des Moines
13	Poster Easel	Easel for Foorm Board	2	$23.52	☑	Amazon	
14	One-page flyers	One-pager with QR code	100	$51.24	☑	Vistaprint	12 in inventory
15	iPads/iPhones/Laptops	iPads - Nikhil, Ashwin iPhones - Ashwin Androids - Nikhil, Henry, Abhu Laptops - Company Mac, Henry, Ashwin, Nikhil	6 + 5 laptops	Free	☑	-	
16	Camera Gear	Henry's stuff	1	Free	☑		
17	QR Code Sign up + Promo code	QR code on fancy design	2	Free	☑	Print Ourself	Remove state-specific information
18	**Giveaways**						
19	Note Cards	Write name, phone #, email on	500	Inventory	☑	OfficeMax	
20	Pen		10	Inventory	☑	OfficeMax	
21	Jar	For dropping names into	1	Inventory	☑	OfficeMax	
22	Instructions for Giveaway (prize menu)	Page of instructions on fancy design	1	Free	☑	Print ourself	Remove state-specific information
23	Prize - Free Year	1 year of Casmium	8 prizes	$800.00	☐		
24	**Refreshments**				☑		
25	Big League Chew Gum	$16 per pack of 12	13	Inventory	☑	Amazon	
26	**Branded Merch**				☑		
27	Quarter-Zip	Fleece/Poly	4	Inventory	☑	VistaPrint	
28	Casmium Stickers	Logos + Support Iowa Baseball	50 logo, 50 baseball, 50 softball	$90.27	☑	VistaPrint	Inventory: 37 logos, 76 Support Iowa Baseball

Figure 5: The team marketing budget with materials, online channels, and event costs

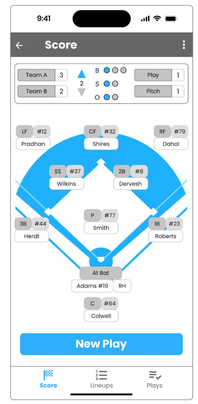


Figure 6: *Stats Tracker* v0.3 with the addition of a diamond screen to manage in-game scoring.

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https://casmium.com

CORE assessment adapted from Middlebrooks, A., Allen, S., McNutt, M., & Morrison, J. (2019). Discovering Leadership: Designing Your Success. Thousand Oaks, CA: Sage Publishers

VUCA assessment adapted from InnovationTraining.org